

Assignment 05 – Literate Programming with Quarto

Tiffany Wickens

8 October 2025

Table of contents

1	Question 1	1
2	Question 2	1
3	Question 3	3
3.1	Key Insights	5
4	Question 4:	5
4.1	Interpration	5
5	Question 5:	7
5.1	Interpretation	7
6	Question 6: Summary and Interpretation	7
7	Question 7	8

1 Question 1

2 Question 2

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```

import wbgapi as wb
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

indicators = {
    'gdp_per_capita': 'NY.GDP.PCAP.CD',
    'gdp_growth_rate': 'NY.GDP.MKTP.KD.ZG',
    'inflation_rate': 'FP.CPI.TOTL.ZG',
    'unemployment_rate': 'SL.UEM.TOTL.ZS',
    'total_population': 'SP.POP.TOTL',
    'life_expectancy': 'SP.DYN.LE00.IN',
    'adult_literacy_rate': 'SE.ADT.LITR.ZS',
    'income_inequality': 'SI.POV.GINI',
    'health_expenditure_gdp_share': 'SH.XPD.CHEX.GD.ZS',
    'measles_immunisation_rate': 'SH.IMM.MEAS',
    'education_expenditure_gdp_share': 'SE.XPD.TOTL.GD.ZS',
    'primary_school_enrolment_rate': 'SE.PRM.ENRR',
    'exports_gdp_share': 'NE.EXP.GNFS.ZS'
}

country_codes = wb.region.members('WLD')
df = wb.data.DataFrame(indicators.values(), economy=country_codes, time=2022, skipBlanks=True)
df = df.drop(columns=['economy'], errors='ignore')
df.rename(columns=lambda x: {v: k for k, v in indicators.items()}.get(x, x).lower(), inplace=True)
df = df.sort_values('country', ascending=True)
df = df.reset_index(drop=True)
print(df.shape)
print(df.head(3))
df.to_csv('wdi.csv', index=False)

```

(217, 14)

	country	inflation_rate	exports_gdp_share	gdp_growth_rate	\
0	Afghanistan	13.712102	18.380042	-6.240172	
1	Albania	6.725203	37.197082	4.826696	
2	Algeria	9.265516	30.808979	3.600000	

	gdp_per_capita	adult_literacy_rate	primary_school_enrolment_rate	\
0	357.261153	NaN	NaN	
1	6846.426694	NaN	96.371230	
2	4961.552577	NaN	105.747154	

	education_expenditure_gdp_share	measles_immunisation_rate	\
--	---------------------------------	---------------------------	---

0	NaN	56.0
1	2.729770	86.0
2	4.749247	79.0

	health_expenditure_gdp_share	income_inequality	unemployment_rate \
0	23.088169	NaN	14.100
1	6.193681	NaN	10.137
2	3.623043	NaN	12.346

	life_expectancy	total_population
0	65.617	40578842.0
1	78.769	2777689.0
2	76.129	45477389.0

3 Question 3

In this section, we conduct exploratory data analysis on three key indicators: GDP per capita, Life expectancy, and Unemployment rate across countries in 2022. We examine their distributions, summary statistics, and potential relationships.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("wdi.csv")
df.head()

#| label: summary-stats
#| echo: true
df[['gdp_per_capita', 'life_expectancy', 'unemployment_rate']].describe()

#| label: correlation
#| echo: true
df[['gdp_per_capita', 'life_expectancy', 'unemployment_rate']].corr()

#| label: distributions
#| fig-cap: "Distribution of Key Indicators (2022)"
#| echo: true
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
sns.histplot(df['gdp_per_capita'], kde=True, ax=axes[0])
axes[0].set_title('GDP per Capita')
```

```

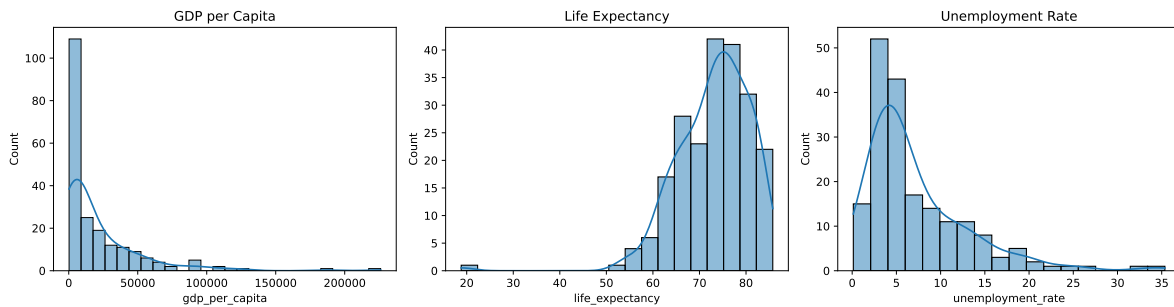
sns.histplot(df['life_expectancy'], kde=True, ax=axes[1])
axes[1].set_title('Life Expectancy')

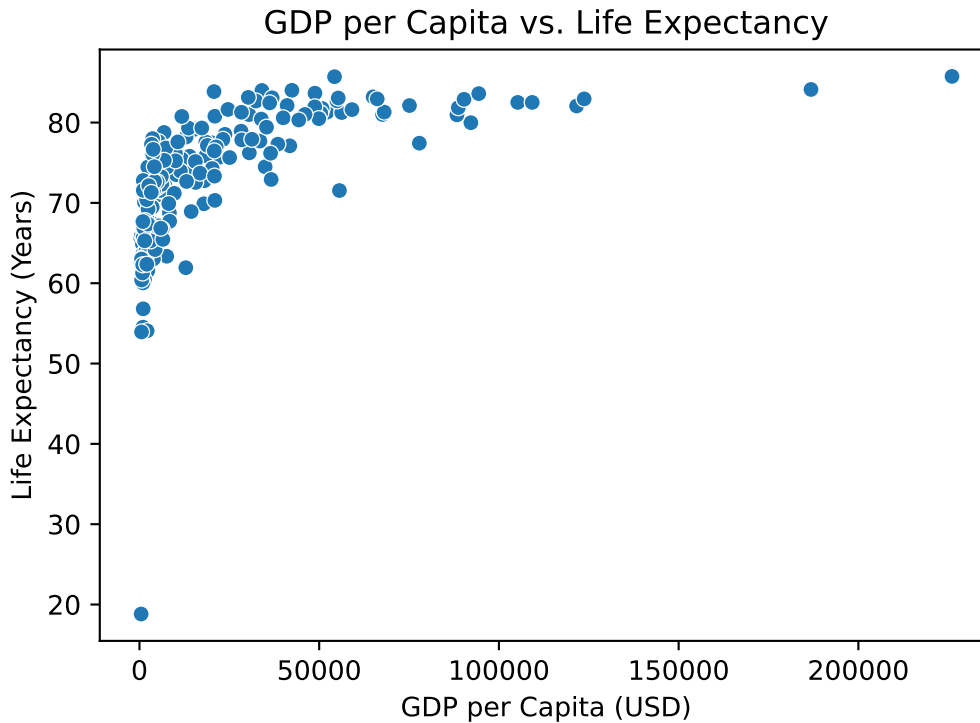
sns.histplot(df['unemployment_rate'], kde=True, ax=axes[2])
axes[2].set_title('Unemployment Rate')

plt.tight_layout()
plt.show()

#| label: scatter-gdp-life
#| fig-cap: "GDP per Capita vs. Life Expectancy (2022)"
#| echo: true
plt.figure(figsize=(6,4))
sns.scatterplot(data=df, x='gdp_per_capita', y='life_expectancy')
plt.title('GDP per Capita vs. Life Expectancy')
plt.xlabel('GDP per Capita (USD)')
plt.ylabel('Life Expectancy (Years)')
plt.show()

```





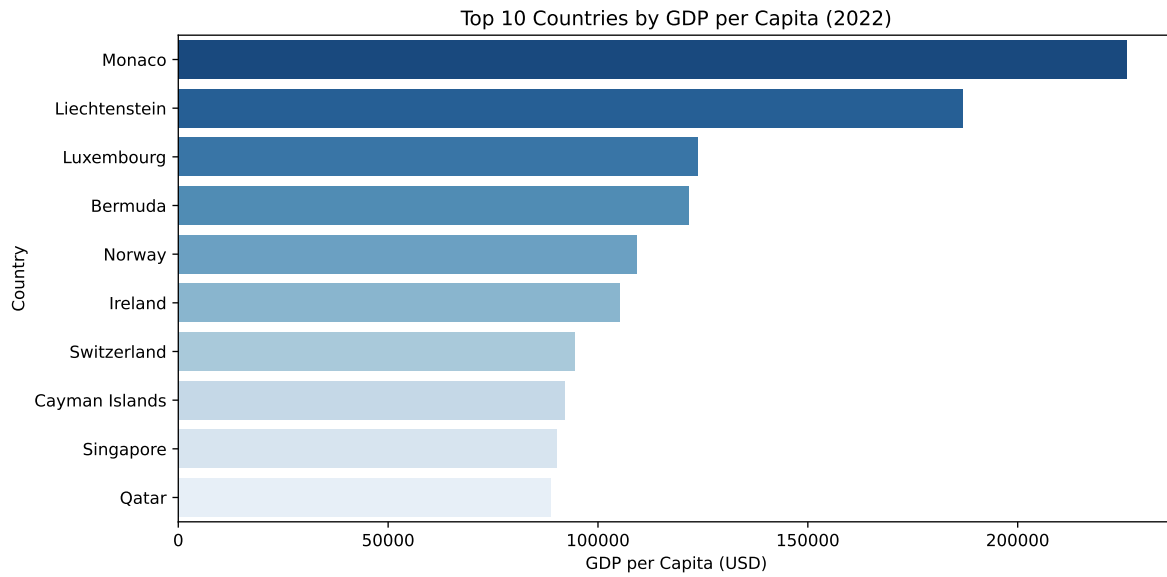
3.1 Key Insights

- **GDP per Capita:** The distribution is right-skewed, with a few high-income countries pulling up the average.
- **Life Expectancy:** Most countries fall between 65–80 years, with higher values correlated with higher GDP per capita.
- **Unemployment Rate** Most countries cluster below 10%, but there are several outliers above 15%.
- **Correlation:** There is a positive relationship between GDP per capita and life expectancy, suggesting wealthier countries tend to have better health outcomes.

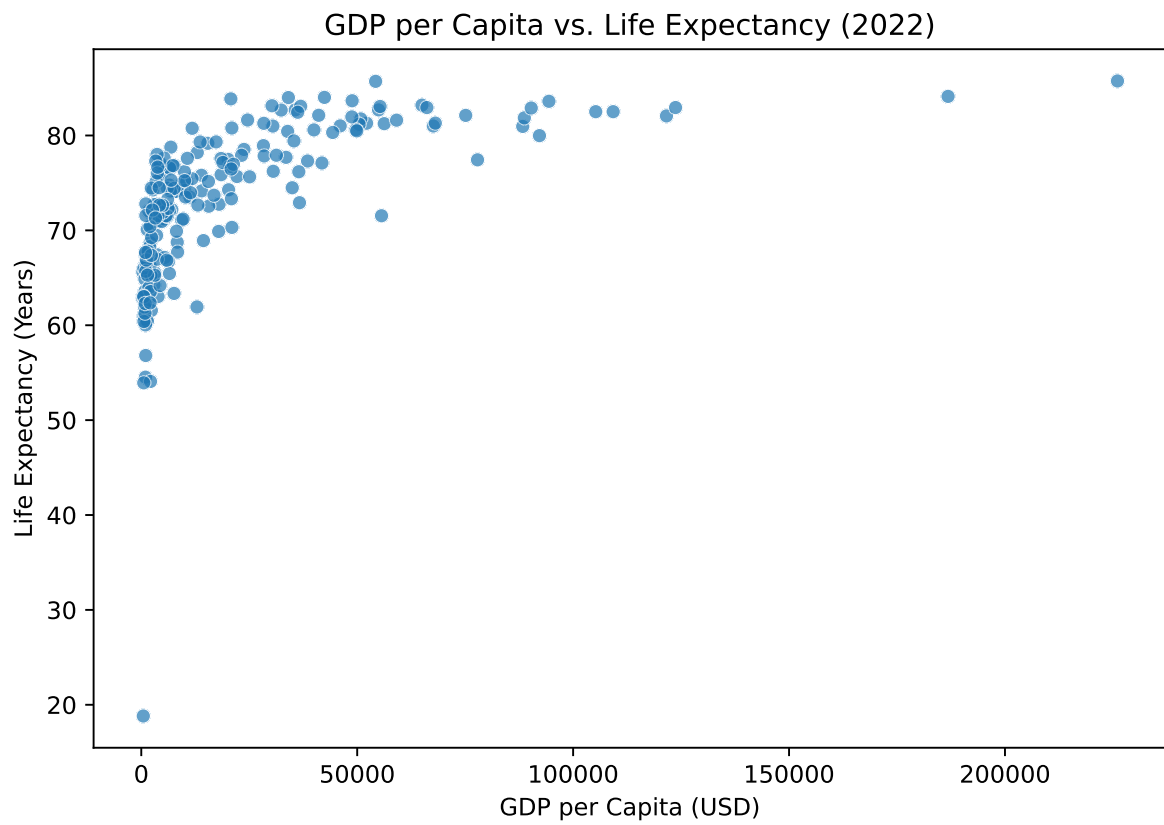
4 Question 4:

4.1 Interpretation

- **Bar Chart:** The top 10 countries by GDP per capita are primarily high-income economies such as Luxembourg, Switzerland, and Ireland.



(a) Top 10 Countries by GDP per Capita (2022). Data source: World Bank.



(b)

Figure 1

- Scatter Plot: The scatter shows a clear positive correlation between GDP per capita and life expectancy, consistent with the general trend that wealthier nations tend to have better health outcomes.

5 Question 5:

Table 1: United States vs. high-income and global averages (GDP per capita, life expectancy, unemployment rate) — World Bank WDI (2022).

Table 1

Indicator	United States	High-income average	Global average
GDP per capita (US\$)	77,860.91	69255.110000	21175.310000
Life expectancy (years)	77.43	81.280000	73.110000
Unemployment rate (%)	3.65	4.640000	7.230000

5.1 Interpretation

The figures show that the United States pairs very high income per person with life expectancy below the high-income average, while its unemployment rate is comparatively low.

- **GDP per capita (US\$):** The U.S. outpaces both high-income peers and the world average by a wide margin.
- **Life expectancy (years):** Despite strong economic capacity, U.S. longevity trails many high-income countries, pointing to system-level inefficiencies and social determinants beyond income alone.
- **Unemployment rate (%):** U.S. unemployment is generally lower than the global average and similar to (or slightly below) the high-income average, reflecting a relatively tight labor market in 2022.

6 Question 6: Summary and Interpretation

Overall Findings (2022):

- **Top performers:** The bar chart in Figure 1 highlights how a handful of small, high-income economies dominate global GDP per capita rankings. This underscores how uneven global wealth remains.
- **Income–health connection:** The scatter plot in ?@fig-gdp-life shows a clear positive relationship between GDP per capita and life expectancy. Nations with higher income levels tend to have longer lifespans, although the slope flattens at the top—suggesting that once basic living standards and healthcare are established, additional income contributes less to longevity.
- **U.S. comparison:** The table in Table 1 shows that the United States stands out for its strong economic performance but below-average life expectancy relative to its peers. It also maintains a comparatively low unemployment

rate, pointing to a strong labor market that does not necessarily translate into better health outcomes.

Interpretation: - Economic strength generally supports better health, but wealth alone is not enough. Differences in healthcare access, social policy, and equity likely explain why some high-income nations achieve higher life expectancy with similar or even lower GDP levels. - A low unemployment rate can reflect labor-market stability and productivity, yet broader quality-of-life outcomes—like public health—depend on how national income is distributed and invested.

Limitations: - These patterns are drawn from a single year (2022) and three indicators. Broader analyses across multiple years or with additional social variables (education, inequality, or environmental data) would give a clearer picture. - Correlations observed in **fig-gdp-life** should not be interpreted as causal—many underlying factors interact to shape both income and health.

Takeaway: The findings reinforce a consistent message: economic resources create potential for improved well-being, but effective policy and equitable systems determine whether that potential is realized. Sustainable prosperity depends as much on how countries allocate and manage resources as on how much they produce.

7 Question 7